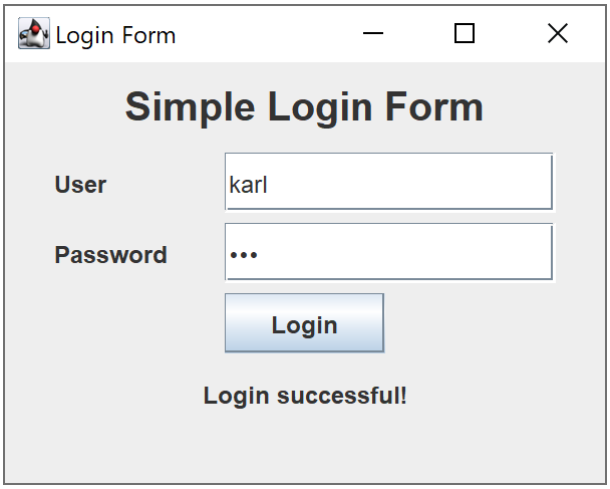
Programming 2

Tutorial 12

Exercise 1 - Build a Login Form

Create a Java Swing application that implements a **Login Form** as shown in this image:



GUI Structure:

The GUI includes a main window (`JFrame`) titled "Login Form". The main windows contains a single container (`JPanel`) sized 300 x 200 which contains all other components.

Following are the components inside the login form:

- Title label: "Simple Login Form"
- Username input (`JTextField`)
- Password input (`JPasswordField`)
- Login button (`JButton`)
- Status label (`JLabel`) for displaying messages

When the user clicks Login, the program should read username and password and validate these against predefined credentials:

- Username: `karl`
- Password: `123`

Upon validation, the GUI should display "Login successful" OR "Login failed".

(*) Notes:

- Use proper naming conventions (`lbl`, `txt`, `btn`, etc.)
- Use proper sizes for containers, labels, text fields, buttons… to make the resulting GUI look as close to the provided image as possible.

- Implement `ActionListener` for event handling.
- Resize the frame to fit the `JPanel` container.

Exercise 2 - Improve Login with Multi-threading

Currently, the login process is instant. In real systems, login may involve long-running blocking operations such as database queries or API calls.

Modify your login form so that:

1. It simulates a slow login process: when user clicks Login, simulate a delay (e.g. 3–5 seconds).
2. Apply Multithreading: the login process must run in a separate thread. The GUI must NOT freeze.
3. Immediately after clicking Login:
    a. Disable the Login button.
    b. Show "Logging in..." in the status label.
4. After processing:
    a. Show success/failure message
    b. Re-enable the button

(*) Hints:

- Use `Thread.sleep()` to simulate delay.
- If your UI freezes after clicking Login, you did it wrong!
- Use the `setEnabled()` method to disable a `JButton`.

Exercise 3 - Display data using `JTable`

Create a window that contains a `JTable` to display data. The table must be placed inside a `JScrollPane`. It has the following columns: ID, Name, Age. Fill it with the following data:

ID	Name	Age
1	Alice	20
2	Bob	22
3	Carol	19

(You may choose your own data)

The table must have appropriate column widths. The user must be able to scroll if data exceeds visible area. You must use a table model to create the table.